

Math 4441: Probability and Statistics

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Chapter 2: Random Variables (Part 2)

Bivariate and Multivariate Random Variables

Lectures 09 – 12

Math 4441: Probability and Statistics

References:

1. Goodman and Yates – Introduction to Probability and Stochastic Processes, 3rd Edition
2. Scott Miller and Donald Childers – Probability and Random Processes, 2nd Edition
3. Bertsekas and Tsitsiklis – Introduction to Probability, 2nd Edition

Introduction

- Chapter 2 (Part 1) explained random experiments with outcome as a single numerical value from a discrete and continuous set.
- The probability models were developed for discrete and continuous random variables
- The probability models include
 - Probability mass function (PMF)
 - Probability density function (PDF)
 - Cumulative distribution function (CDF)

- We will study random experiments with two numerical values as outcomes
 - Both from discrete set (Bivariate discrete distribution)
 - Both from continuous set (Bivariate continuous distribution)
 - One from a discrete set and the other from a continuous set (Bivariate mixed distribution)
- For example, in the classification of a set transmitted and received signals, each signal can be classified as – high, medium and low quality
- Random values might represent
 - Number of high-quality signals received
 - Number of low-quality signals received
- Accordingly, we can use random variables X and Y to represent the random values

The simultaneous randomness of X and Y or the joint probability distribution of them will represent the probability model

- Similarly, two random values from two continuous sets might be associated with a single random experiment,
 - Need probability models for simultaneous behavior of two continuous random variables, X and Y .

Finally, a bivariate mixed distribution deals with probability models associated with one discrete and one continuous random variable

Bivariate discrete distributions

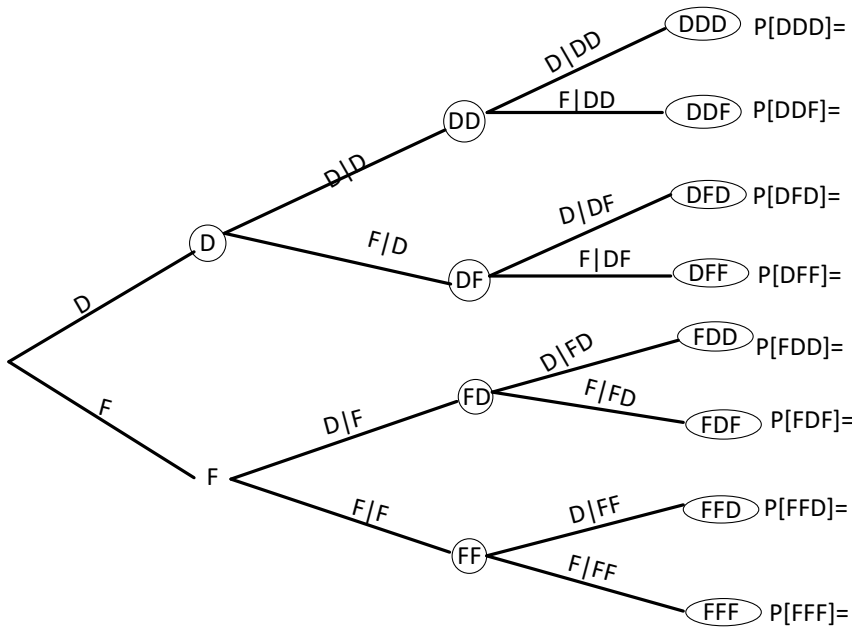
A bivariate discrete distribution is associated for random experiments with

- A pair of discrete random variables
- Joint discrete random variables
- Bivariate discrete random variables

And the probability models include:

- Joint probability mass function (JMPF)
 - Marginal probability mass function (MPMF)
- Joint cumulative distribution function (JCDF)
 - Marginal cumulative distribution function (MCDF)

Example: Send three packets from one computer to another



Sample space of the experiment

$$S = \{FFF, FFD, FDF, FDD, DFF, DFD, DDF, DDD\}$$

Random variables:

X : # of successful deliveries

$$S_X = \{0, 1, 2, 3\}$$

Y : # of successful deliveries before any failure occurs

$$S_Y = \{0, 1, 2, 3\}$$

- Each outcome specifies a pair of values of X and Y .
- Single random variable: an outcome represents a point on the number line
- Bivariate random variables: an outcome represents a point on the 2-D plane
- Let $g(s)$ be the function that transform each outcome into a pair of random values. Then

$$g(DDD) = (3, 3), \quad g(DDF) = (2, 2), \quad g(DFD) = (2, 1), \quad g(DFF) = (1, 1)$$

$$g(FDD) = (2, 0), \quad g(FDF) = (1, 0), \quad g(FFD) = (1, 0), \quad g(FFF) = (0, 0)$$

- Observing the tree diagram, we have

$$P[DDD] = \frac{27}{64}, P[DDF] = \frac{9}{64}, P[DFD] = \frac{9}{64}, P[DFF] = \frac{3}{64}, P[FDD] = \frac{9}{64},$$

$$P[FDF] = \frac{3}{64}, P[FFD] = \frac{3}{64}, P[FFF] = \frac{1}{64}$$

The probability model of a single discrete random variable represents

- Set of real numbers S_X – each representing an event – a function $g(\cdot)$ converts each outcome into a real number
- Real numbers are points on the number line
- Probability of occurring each point (a number) as the outcome of the experiment – probability of occurring the event $\{X = x\}$ or $\{X \leq x\}$

The probability model of a pair of random variables represents

- Set of pair of real numbers S_{XY} – each representing an event – a function $g(\cdot)$ converts each outcome into a pair of real numbers
- Pair of real numbers are points on a 2-D plane
- Probabilities of occurring each point (a pair of numbers) as the outcome of the experiment – probability of occurring the event $\{X = x\} \cap \{Y = y\}$ or $\{X \leq x\} \cap \{Y \leq y\}$

The set of pairs of simultaneous values of X and Y ,

a point (x, y) on the 2-D plane,

Can be given by

$$S_{XY} = \{(3, 3), (2, 2), (2, 1), (2, 0), (1, 1), (1, 0), (0, 0)\}$$

The probabilities of the pairs of values (or the points) as the outcomes are:

$$P[(3, 3)] = P[X = 3, Y = 3] = P[DDD] = 27/64$$

$$P[(2, 2)] = P[X = 2, Y = 2] = P[DDF] = 9/64$$

$$P[(2, 1)] = P[X = 2, Y = 1] = P[DFD] = 9/64$$

$$P[(2, 0)] = P[X = 2, Y = 0] = P[FDD] = 9/64$$

$$P[(1, 1)] = P[X = 1, Y = 1] = P[DFF] = 3/64$$

$$P[(1, 0)] = P[X = 1, Y = 0] = P[FFD] + P[FDF] = 6/64$$

$$P[0, 0] = P[X = 0, Y = 0] = P[FFF] = 1/64$$

Joint Probability Mass Function (JPMF)

Definition 4.1 (Joint Probability Mass Function). The joint probability mass function of two discrete random variables X and Y , denoted as $P_{XY}(x, y)$, is defined as

$$P_{XY}(x, y) = P[X = x, Y = y],$$

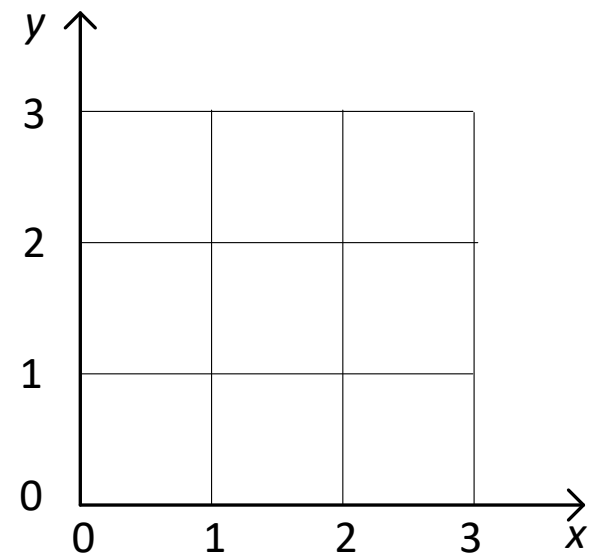
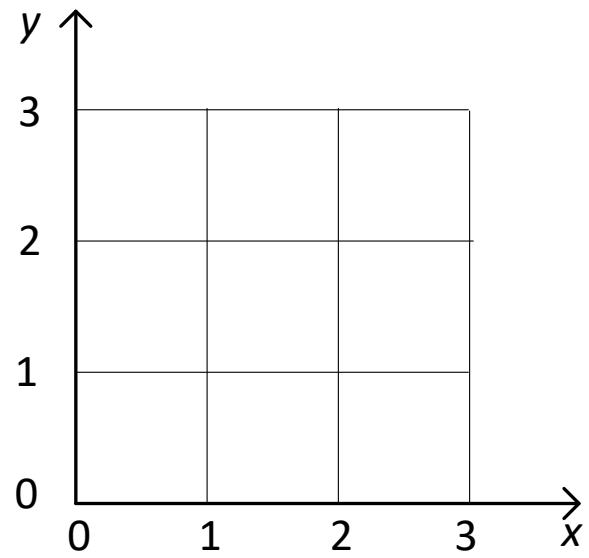
and it satisfies,

$$\begin{aligned} P_{XY}(x, y) &\geq 0; \\ \sum_x \sum_y P_{XY}(x, y) &= 1 \end{aligned}$$

The PMF of a single discrete random variable is represented as a list,

The JPMF of a pair of random variables can be represented in a number of ways.

The bivariate PMF or the JPMF can be represented as a graph



The bivariate PMF or JPMF can be represented as a list as well

$$P_X(x) = \begin{cases} 27/64, & x = 3, y = 3 \\ 9/64, & x = 2, y = 2 \\ 9/64, & x = 2, y = 1 \\ 9/64, & x = 2, y = 0 \\ 3/64, & x = 1, y = 1 \\ 6/64, & x = 1, y = 0 \\ 1/64, & x = 0, y = 0 \\ 0, & \text{otherwise} \end{cases}$$

A third presentation of the joint PMF is the matrix given below:

$P_{XY}(x,y)$	$y = 0$	$y = 1$	$y = 2$	$y = 3$	
$x = 0$					
$x = 1$					
$x = 2$					
$x = 3$					

Summary:

1. Bivariate distributions are used for experiments with two outcomes
2. Random variable X represents one value along the x -axis and Y represents the other value along the y -axis
3. Outcomes are an ordered-pair (x, y) and represents a point on the 2-D plane
4. JPMF of X and Y is defined as
$$P_{XY}(x, y) = P[X = x, Y = y]$$

And represents the probability of simultaneously occurring the value of x and y , i.e., occurring the even $\{X = x\} \cap Y = y\}$
5. Marginal PMFs of X and Y are given by

$$P_X(x) = \sum_y P_{XY}(x, y) \text{ and } P_Y(y) = \sum_x P_{XY}(x, y)$$

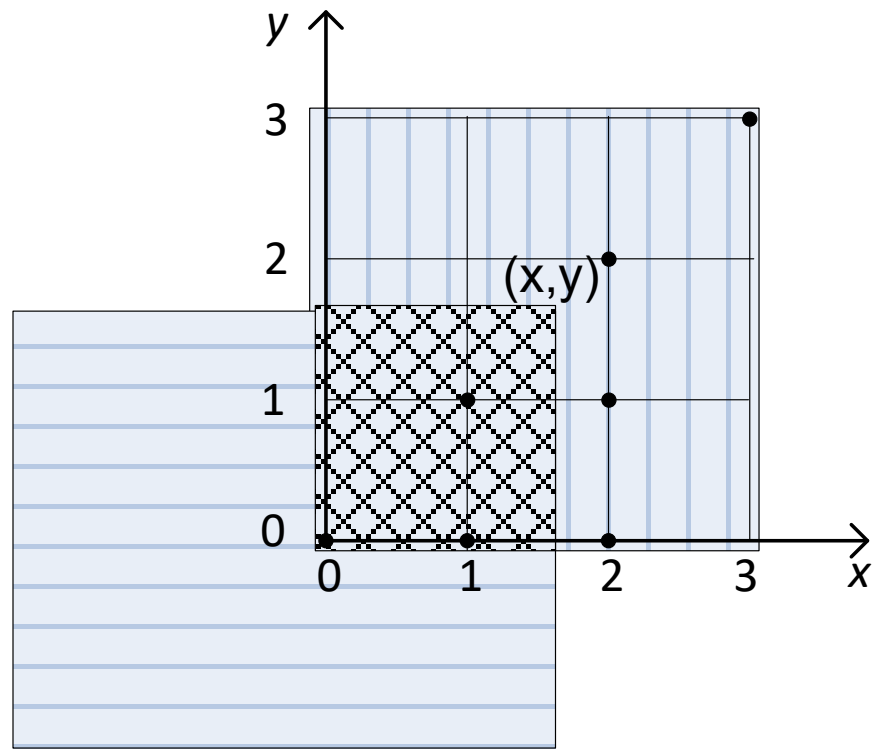
$P_{XY}(x, y)$	$y = 0$	$y = 1$	$y = 2$	$y = 3$	$P_X(x)$
$x = 0$	1/64	0	0	0	1/64
$x = 1$	6/64	3/64	0	0	9/64
$x = 2$	9/64	9/64	9/64	0	27/64
$x = 3$	0	0	0	27/64	27/64
$P_Y(y)$	16/64	12/64	9/64	27/64	1

Joint Cumulative Distribution Function

Definition 4.5 (JCDF). The joint cumulative distribution function (JCDF) of random variables X and Y , denoted as $F_{XY}(x, y)$, is defined as

$$F_{XY}(x, y) = P[X \leq x \cap Y \leq y] = P[X \leq x, Y \leq y]$$

for all x and y . It is the probability of the intersection of two events: random variables X and Y therefore induce a probability distribution over a two-dimensional Euclidean plane.



$F_{XY}(x, y)$ has the following properties:

1. $F_{XY}(-\infty, -\infty) = F_{XY}(x, -\infty) = F_{XY}(-\infty, y) = 0.$

2. $F_{XY}(+\infty, +\infty) = 1.$

3.

$$F_{XY}(x, y) = \sum_{i=-\infty}^x \sum_{j=-\infty}^y P_{XY}(i, j).$$

any x_1, x_2, y_1 and y_2 such that $x_1 < x_2$ and $y_1 < y_2$, the probability $P[x_1 < X \leq x_2, y_1 < Y \leq y_2]$ is given in terms of $F_{XY}(x, y)$ by

$$P[x_1 < X \leq x_2, y_1 < Y \leq y_2] = F_{XY}(x_2, y_2) - F_{XY}(x_1, y_2) - F_{XY}(x_2, y_1) + F_{XY}(x_1, y_1).$$

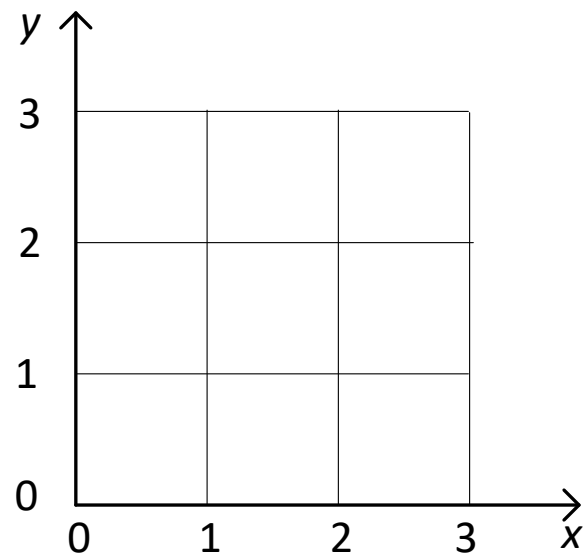
How to represent the JCDF

Single Random Variable:

The CDF $F_X(x)$ represents –

The sum of the PMFs of all points on the number line that are

On the line $X = x$, or to the left of the line $X = x$



Bivariate Random Variable:

The JCDF $F_{XY}(x, y)$ represents –

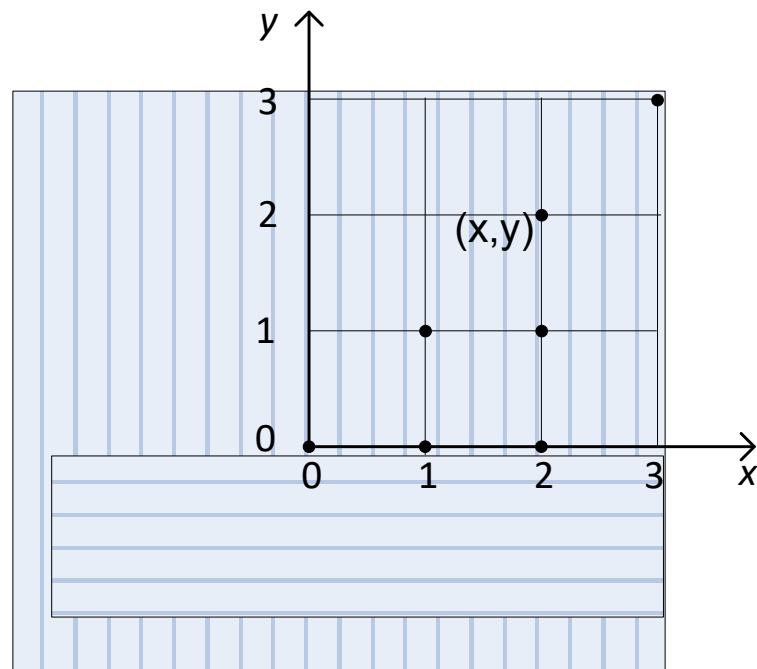
The sum of the JPMFs of all points on the 2-D plane that satisfy following two conditions:

1. On the line $X = x$, or to the left of the line $X = x$
2. On the line $Y = y$, or below the line $Y = y$

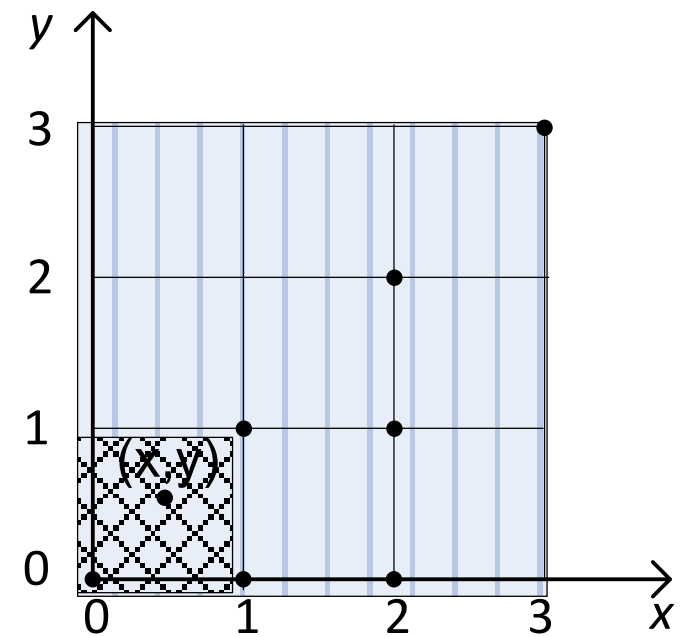
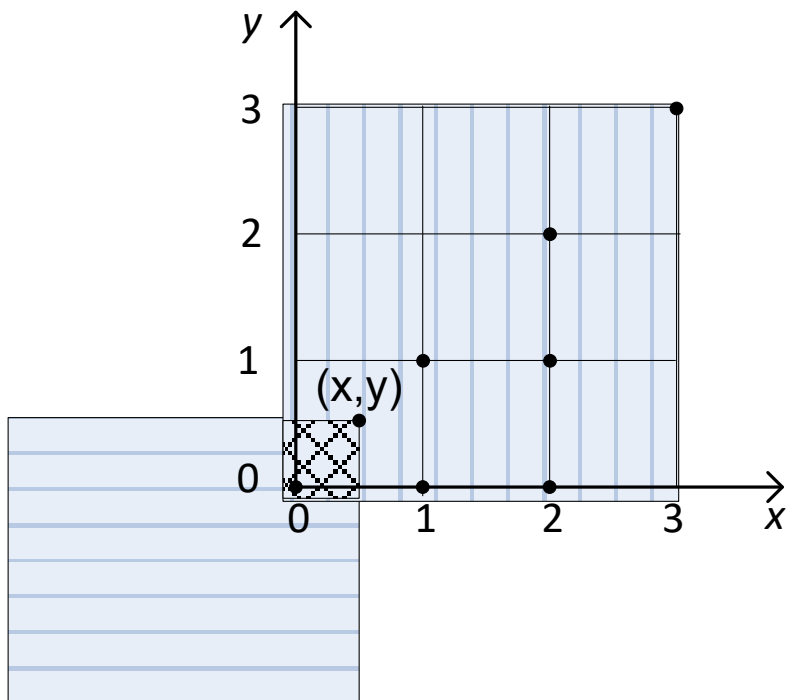
JCDF for random variable X and Y with JPDF given by

$$P_X(x) = \begin{cases} 27/64, & x = 3, y = 3 \\ 9/64, & x = 2, y = 2 \\ 9/64, & x = 2, y = 1 \\ 9/64, & x = 2, y = 0 \\ 3/64, & x = 1, y = 1 \\ 6/64, & x = 1, y = 0 \\ 1/64, & x = 0, y = 0 \\ 0, & \text{otherwise} \end{cases}$$

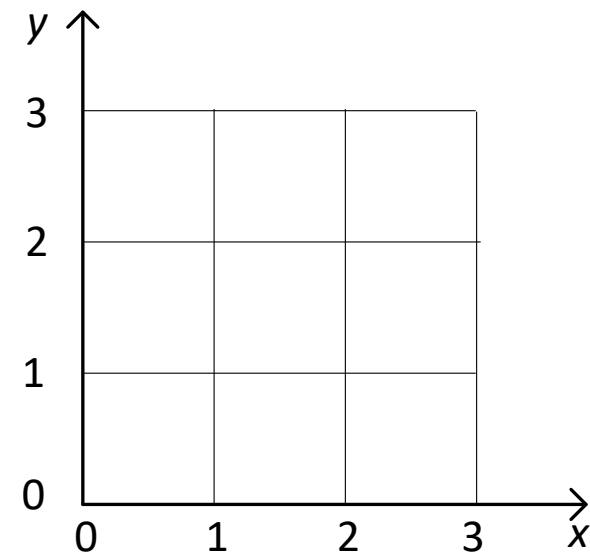
1. For $X < 0$ OR $Y < 0$



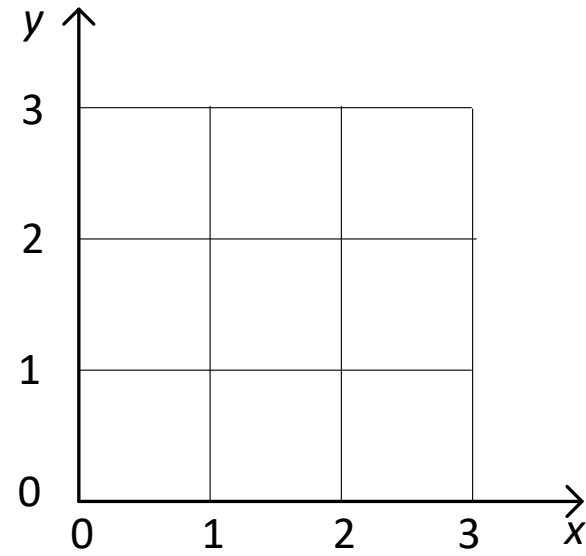
2. For $0 \leq X < 1$ AND $Y \geq 0$



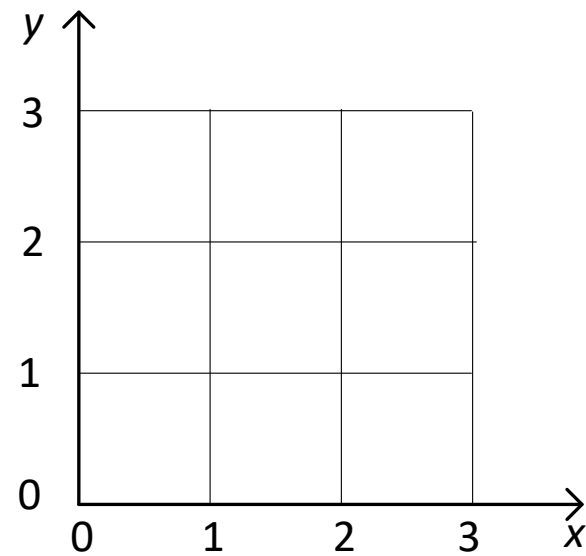
3. For $1 \leq X < 2$ AND $0 \leq Y < 1$



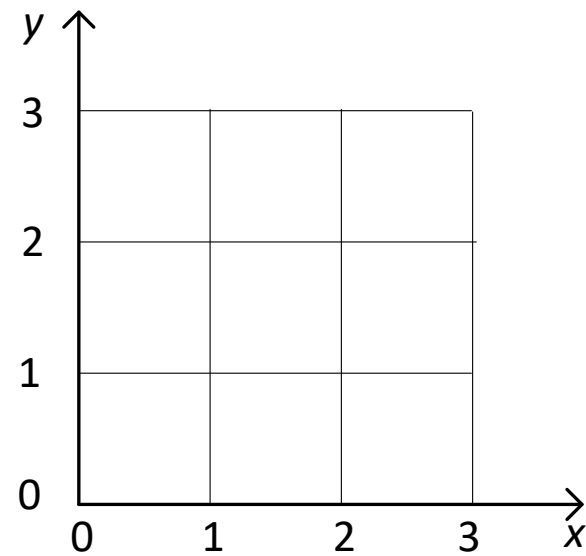
4. For $1 \leq X < 2$ AND $Y \geq 1$



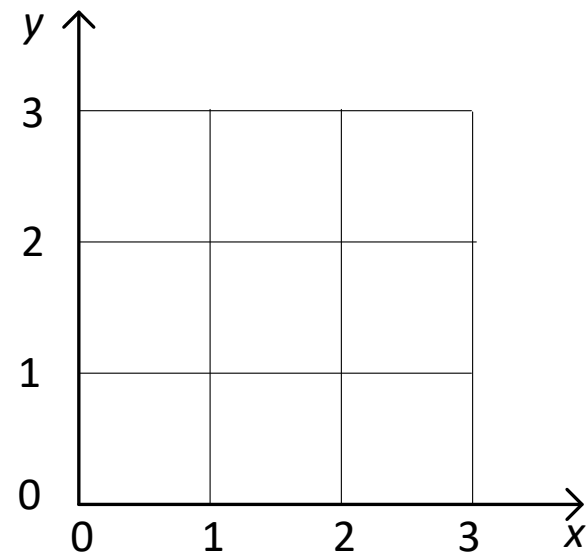
5. For $x \geq 2$ AND $0 \leq y < 1$



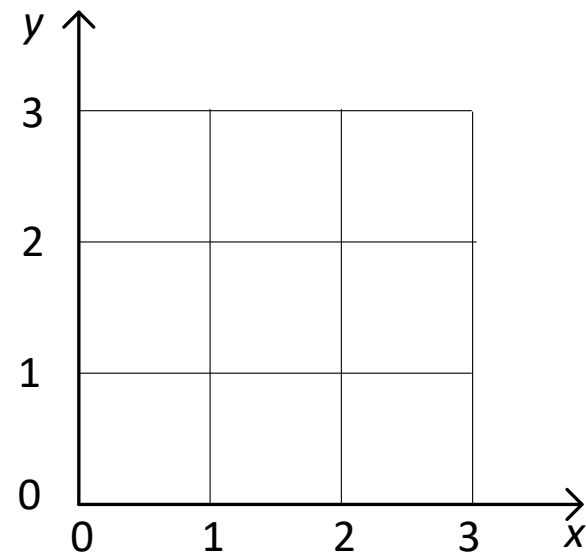
6. For $x \leq 2$ AND $1 \leq y < 2$



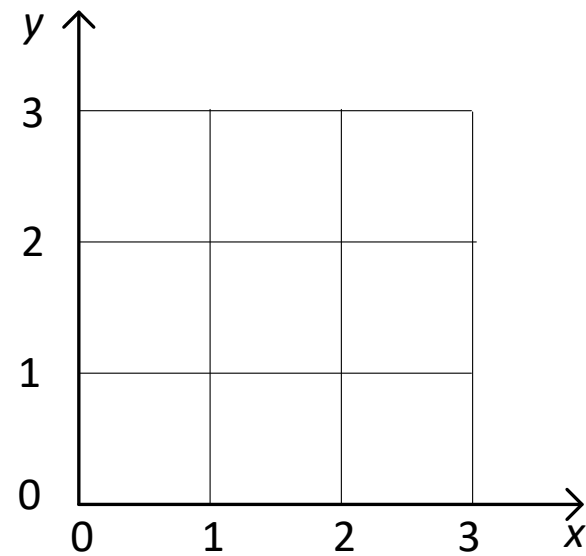
7. For $x \geq 2$ AND $2 \leq y < 3$



8. For $2 \leq x < 3$ AND $y \geq 3$



9. For $x \geq 3$ AND $y \geq 3$



Bivariate discrete probability models (JPMF and JCDF)

Example: Send 3 packets from one computer to another

Sample space $S = \{FFF, FFD, FDF, FDD, DFF, DFD, DDF, DDD\}$

Random variables:

X : # of successful deliveries $S_X = \{0, 1, 2, 3\}$

Y : # of successful deliveries before any failure $S_Y = \{0, 1, 2, 3\}$

Function that converts each $\omega \in S$ into a pair of real numbers

$g(DDD) = (3, 3), \quad g(DDF) = (2, 2), \quad g(DFD) = (2, 1), \quad g(DFF) = (1, 1)$
 $g(FDD) = (2, 0), \quad g(FDF) = (1, 0), \quad g(FFD) = (1, 0), \quad g(FFF) = (0, 0)$

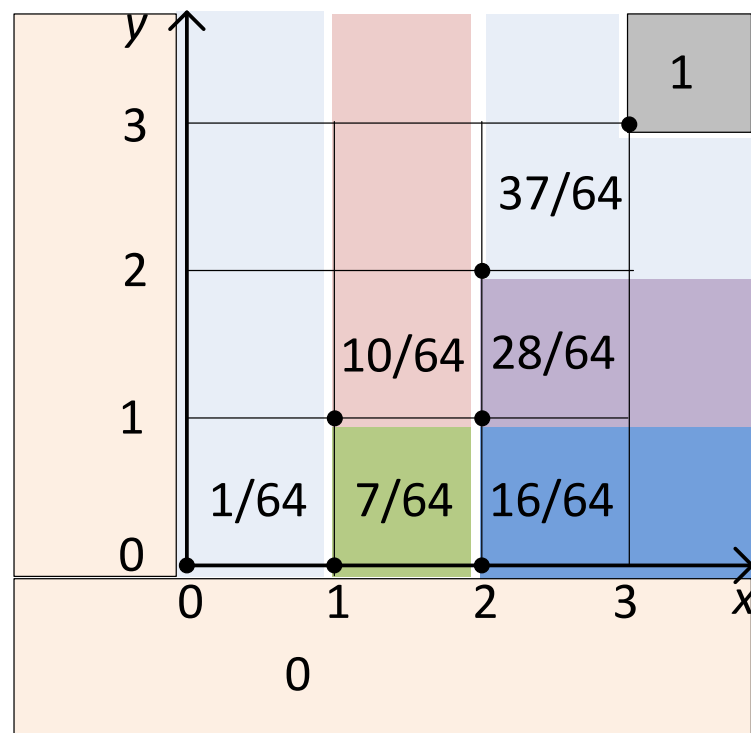
Probability associated with each pair of real numbers

$$P\{DDD\} = \frac{27}{64}, P\{DDF\} = \frac{9}{64}, P\{DFD\} = \frac{9}{64}, P\{DFF\} = \frac{3}{64}, P\{FDD\} = \frac{9}{64},$$

$$P\{FDF\} = \frac{3}{64}, P\{FFD\} = \frac{3}{64}, P\{FFF\} = \frac{1}{64}$$

$$P_X(x) = \begin{cases} 27/64, & x = 3, y = 3 \\ 9/64, & x = 2, y = 2 \\ 9/64, & x = 2, y = 1 \\ 9/64, & x = 2, y = 0 \\ 3/64, & x = 1, y = 1 \\ 6/64, & x = 1, y = 0 \\ 1/64, & x = 0, y = 0 \\ 0, & \text{otherwise} \end{cases}$$

$P_{XY}(x, y)$	$y = 0$	$y = 1$	$y = 2$	$y = 3$	$P_X(x)$
$x = 0$	1/64	0	0	0	1/64
$x = 1$	6/64	3/64	0	0	9/64
$x = 2$	9/64	9/64	9/64	0	27/64
$x = 3$	0	0	0	27/64	27/64
$P_Y(y)$	16/64	12/64	9/64	27/64	1



$$F_X(x) = \begin{cases} 0, & x < 0 \text{ OR } y < 0 \\ \frac{1}{64}, & 0 \leq x < 1, y \geq 0 \\ \frac{7}{64}, & 1 \leq x < 2, 0 \leq y < 1 \\ \frac{10}{64}, & 1 \leq x < 2, y \geq 1 \\ \frac{16}{64}, & x \geq 2, 0 \leq y < 1 \\ \frac{28}{64}, & x \geq 2, 1 \leq y < 2 \\ \frac{37}{64}, & x \geq 2, 2 \leq y < 3 \\ & 2 \leq x < 3, y \geq 3 \\ 1, & x \geq 3, y \geq 3 \end{cases}$$

Properties of JPMF

1. $P_{XY}(x, y) \geq 0$

2. $\sum_x \sum_y P_{XY}(x, y) = 1$

3. $P_X(x) = \sum_y P_{XY}(x, y)$

4. $P_Y(y) = \sum_x P_{XY}(x, y)$

Properties of JCDF

$F_{XY}(x, y)$ has the following properties:

1. $F_{XY}(-\infty, -\infty) = F_{XY}(x, -\infty) = F_{XY}(-\infty, y) = 0.$

2. $F_{XY}(+\infty, +\infty) = 1.$

3.

$$F_{XY}(x, y) = \sum_{i=-\infty}^x \sum_{j=-\infty}^y P_{XY}(i, j).$$

Marginal Cumulative Distribution Function

Definition 4.6. If X and Y are two discrete random variables with joint CDF $F_{XY}(x, y)$, then marginal CDF of X and Y are

$$\begin{aligned} F_X(x) &= P[X \leq x] \\ &= P[X \leq x \cap Y \leq +\infty] \\ &= F_{XY}(x, \infty). \end{aligned}$$

The above relation follows from the fact that the joint event $X \leq x \cap Y \leq \infty$ is the same as the event $X \leq x$, since $Y \leq \infty$ is a sure event. Similarly,

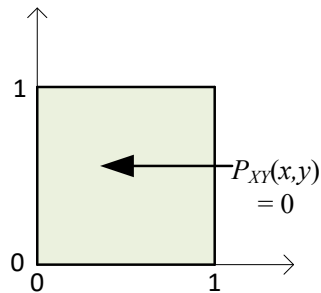
$$F_Y(y) = F_{XY}(\infty, y).$$

Bivariate Continuous Random Variables

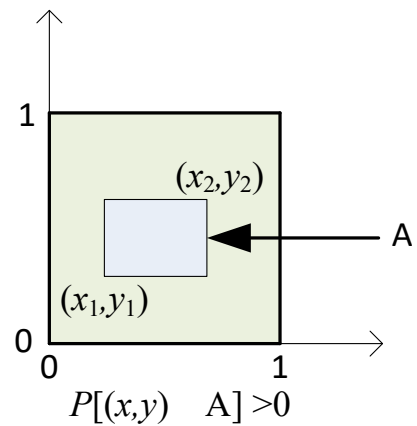
1. Like the PMF of a single continuous random variable is meaningless
 - a. JPMF for joint continuous RVs is meaningless as well
2. JCDF for bivariate continuous RVs are well-defined
3. JPDP for bivariate continuous RVs are also well-defined
4. Marginal PDF and Marginal CDF are used as well

Joint Cumulative Distribution Function (JCDF)

Consider 2 continuous random variables X and Y jointly continuous within a unit square



$$P[X = x, Y = y] = \frac{1}{n} \Big|_{n=\infty} = 0.$$



$$A = [x_1 \leq X \leq x_2, y_1 \leq Y \leq y_2] \in \mathcal{F}$$

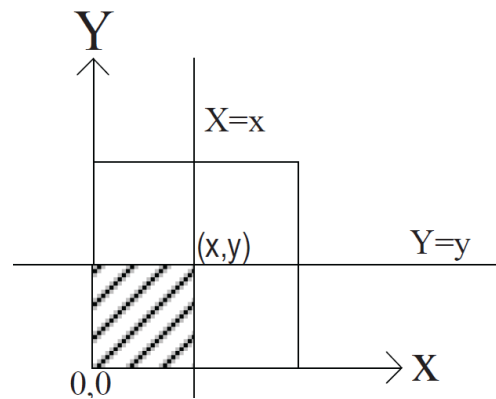
$$P[(x, y) \in A] > 0 \text{ if } 0 \leq x \leq 1, 0 \leq y \leq 1$$

$$P[(x, y) \in A] = P[x_1 \leq X \leq x_2, y_1 \leq Y \leq y_2]$$

$$= \frac{\text{Area of the region } A}{\text{Total area of the square}} = (x_2 - x_1)(y_2 - y_1)$$

When $x_1 = -\infty, x_2 = x, y_1 = -\infty, y_2 = y$,

$$\begin{aligned} P[x_1 \leq X \leq x_2, y_1 \leq Y \leq y_2] &= P[-\infty \leq X \leq x, -\infty \leq Y \leq y] \\ &= P[X \leq x, Y \leq y] = F_{XY}(x, y) \end{aligned}$$



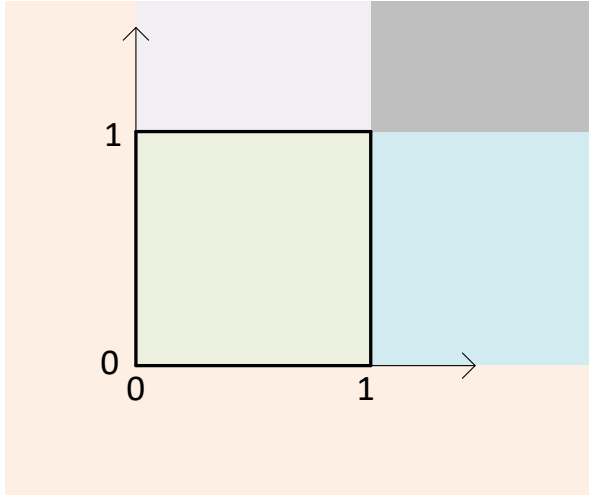
Definition 4.7. The joint cumulative distribution function (JCDF) of a pair of continuous random variables X and Y , defined on a single sample space, is denoted by $F_{XY}(x, y)$, and defined by

$$F_{XY}(x, y) = P[X \leq x, Y \leq y],$$

and it represents the joint probability of occurring two events $\{X \leq x\}$ and $\{Y \leq y\}$ simultaneously.

Mathematical Representation of the CDF

1. The joint CDF of a pair of continuous RVs has to be defined for all points on the $X - Y$ Plane.
2. The whole needs to be divided into sub-regions such that for each sub-region, a single function can define the CDF



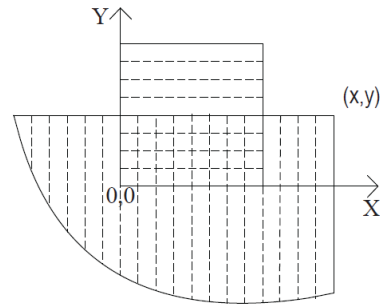
How to calculate the JCDF for a pair of continuous RVs

1. Let X and Y are uniformly distributed within a unit square
2. CDF of the point (x, y) , $F_{XY}(x, y)$ is to be calculated
3. Let S_{XY} denote the area for which X and Y are defined

$$S_{XY} = [0 \leq X \leq 1, 0 \leq Y \leq 1]$$

4. JCDF for (x, y) defines the probability for a region, say R ,

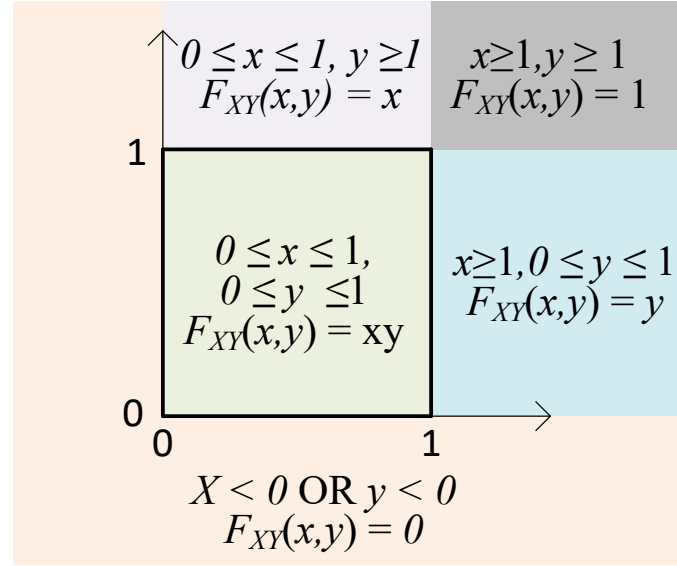
$$R = [-\infty < X \leq x, -\infty < Y \leq 1]$$



5. JCDF defines the probability for the intersection of two regions, $S_{XY} \cap R$

$$S_{XY}(x, y) = S_{XY} \cap R$$

$$F_{XY}(x, y) = P[X \leq x, Y \leq y] = P[(x, y) \in S_{XY}(x, y)]$$



1. Whenever either of the random variables is less than zero, the intersection of the area by the regions $[-\infty < X \leq x, -\infty < Y \leq y]$ and $[0 \leq X \leq 1, 0 \leq Y \leq 1]$ is empty. Therefore, if $X < 0$ or $Y < 0$ then the JCDF is zero, i.e.,

$$F_{X,Y}(x, y) = 0, \quad \text{if } x < 0 \quad \text{or} \quad y < 0.$$

The JCDF of any point where $x > 1$ and $0 \leq y < 1$ will include the double shared region with area $r_2 = y$ and the JCDF is given by

$$F_{X,Y}(x, y) = y \quad \text{if } x > 1 \quad \text{or} \quad 0 \leq y < 1.$$

Any point within the shaded region will have a JCDF given by

$$F_{X,Y}(x, y) = x \quad \text{if } y > 1 \quad \text{or} \quad 0 \leq x < 1.$$

The JCDF of any point (x, y) within the unit square can be represented by a single function. The JCDF is the area of the intersection of the unit square $[0 \leq x \leq 1, 0 \leq y \leq 1]$ and area covered by $F_{X,Y}(x, y)$ which is $[-\infty < X \leq x, -\infty \leq Y \leq y]$, which is $r_3 = x_4$ and the JCDF is given by

$$F_{X,Y}(x, y) = xy \quad \text{if } 0 \leq x \leq 1, \quad 0 \leq y \leq 1.$$

The JCDF of any point within the shaded region can be represented by a single function $r_5 = 1$ and the JCDF is given by

$$F_{X,Y}(x, y) = 1 \quad \text{if } x > 1, \quad y > 1.$$

Finally, the complete JCDF is given by

$$F_{X,Y}(x,y) = \begin{cases} 0, & \text{if } x < 0 \text{ or } y < 0; \\ y, & \text{if } x > 1, 0 \leq y \leq 1; \\ x, & \text{if } 0 \leq x \leq 1, y > 1; \\ xy, & \text{if } 0 \leq x \leq 1, 0 \leq y \leq 1; \\ 1, & \text{if } x > 1, y > 1 \end{cases}$$

Properties of JCDF

- i) The joint cumulative distribution function is a measure of probability, it must take on a value between 0 and 1, i.e.,

$$0 \leq F_{X,Y}(x, y) \leq 1$$

- ii) The random variables X and Y must assume real value. It is impossible for either of these two to take a value less than or equal to $-\infty$, and both must be less than $+\infty$. Therefore,

$$F_{X,Y}(-\infty, y) = 0$$

$$F_{X,Y}(x, -\infty) = 0$$

$$F_{X,Y}(-\infty, -\infty) = 0$$

$$F_{X,Y}(+\infty, +\infty) = 1$$

iii) The probability that X and Y have values within certain intervals can be calculated from their JCDF as shown in Fig. -.- and is given by

$$F_{X,Y}(x_1 \leq X < x_2, y_1 \leq Y < y_2) = F_{X,Y}(x_2, y_2) - F_{X,Y}(x_1, y_2) - F_{X,Y}(x_2, y_1) + F_{X,Y}(x_1, y_1).$$

Joint Probability Density Function

The joint probability density function (JPDF) of jointly distributed random variables X and Y at point (x, y) is denoted by $f_{XY}(x, y)$ and is given by

$$f_{XY}(x, y) =$$

$$f_{X,Y}(x, y) = \begin{cases} 1, & \text{for } 0 < x < 1, 0 < y < 1; \\ 0, & \text{otherwise.} \end{cases}$$

Properties of JPDF

- i. Since the JPDF represents a probability (i.e., probability per unit area), it must be nonnegative.

$$f_{X,Y}(x,y) \geq 0.$$

- ii. Since the JPDF represents probability per unit area, the JPDF multiplied by the total area will give the total probability which is 1.

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_{X,Y}(x,y) dy dx = 1.$$

- iii. An extension of the above property

$$F_{X,Y}(x,y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(x,y) dy dx$$

iv. The probability of getting the pair of values in any bounded region is

$$P[x_1 < X \leq x_2, y_1 < Y \leq y_2] = \int_{x_1}^{x_2} \int_{y_1}^{y_2} f_{X,Y}(x, y) dy dx$$

v. If the two random variables are jointly uniformly distributed over the whole area associated with the random variables, then the JPDF is a constant.

$$f_{X,Y}(x, y) = c.$$

Example 4.6. Random variable X and Y have the joint PDF given by

$$f_{X,Y}(x,y) = \begin{cases} c, & \text{for } 0 \leq x \leq 5, 0 \leq y \leq 3; \\ 0, & \text{otherwise.} \end{cases}$$

- a) Find the value of the constant c .
- b) Find the probability $P[A] = P[2 \leq X \leq 3, 1 \leq Y \leq 3]$.
- c) Find the probability $P[B] = P[Y \geq X]$.

$$\int_0^5 \int_0^3 f_{X,Y}(x,y) dy dx = 1$$

$$\Rightarrow \int_0^5 \int_0^3 c dy dx = 1$$

$$\Rightarrow \int_0^5 \left[cy \Big|_0^3 \right] dx = 1$$

$$\Rightarrow \int_0^5 3c dx = 1$$

$$\Rightarrow 3cx \Big|_0^5 = 3c(5 - 0) = 1$$

$$\Rightarrow 15c = 1$$

$$\Rightarrow c = \frac{1}{15}.$$

$$\begin{aligned}
 P[A] &= \int_2^3 \int_1^3 \frac{1}{15} dy dx \\
 &= \int_2^3 \left[\frac{y}{15} \Big|_1^3 \right] dx \\
 &= \int_2^3 \frac{3-1}{15} dx \\
 &= \frac{2x}{15} \Big|_2^3 \\
 &= \frac{2}{15}(3-2) \\
 &= \frac{2}{15}.
 \end{aligned}$$

Oppositely, the same probability can be found by calculating the ratio of the desired area to the total area. The area of the desired region is $A = 1 \times 2 = 2$ and the total area is $5 \times 3 = 15$. Therefore, the probability is

$$P[A] = \frac{\text{Area of } A}{\text{Total Area}} = \frac{2}{3}.$$

$$P[B] = \int_0^3 \int_x^3 \frac{1}{15} dy dx$$

$$= \int_0^3 \left. \frac{y}{15} \right|_x^3 dx$$

$$= \int_0^3 \frac{3-x}{15} dx$$

$$= \left. \frac{3}{15} x \right|_0^3 - \left. \frac{1}{30} x^2 \right|_0^3$$

$$= \frac{9}{15} - \frac{9}{30}$$

$$= \frac{9}{15} - \frac{3}{5}$$

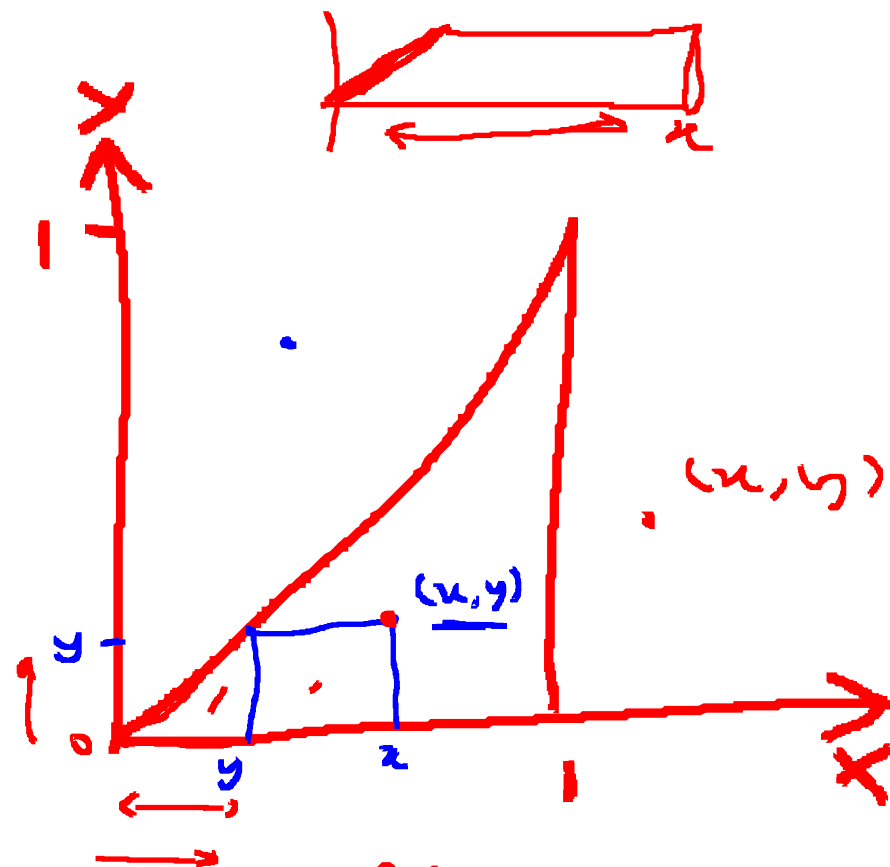
$$\approx 0.1$$

Example 7. Continuous random variables X and Y are defined to be jointly uniform within the region $0 \leq y \leq x \leq 1$.

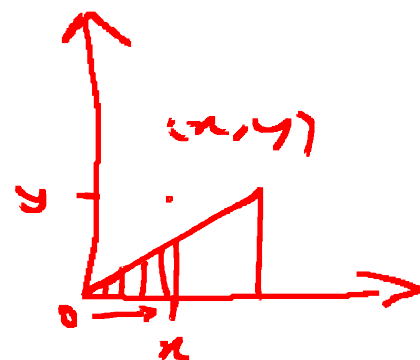
- Find the joint PDF of X and Y , $f_{XY}(x, y)$.
- Find the joint CDF of X and Y , $F_{XY}(x, y)$.

$$\int_0^1 \int_0^x c \, dy \, dx$$

$$\int_0^y \int_v^y$$

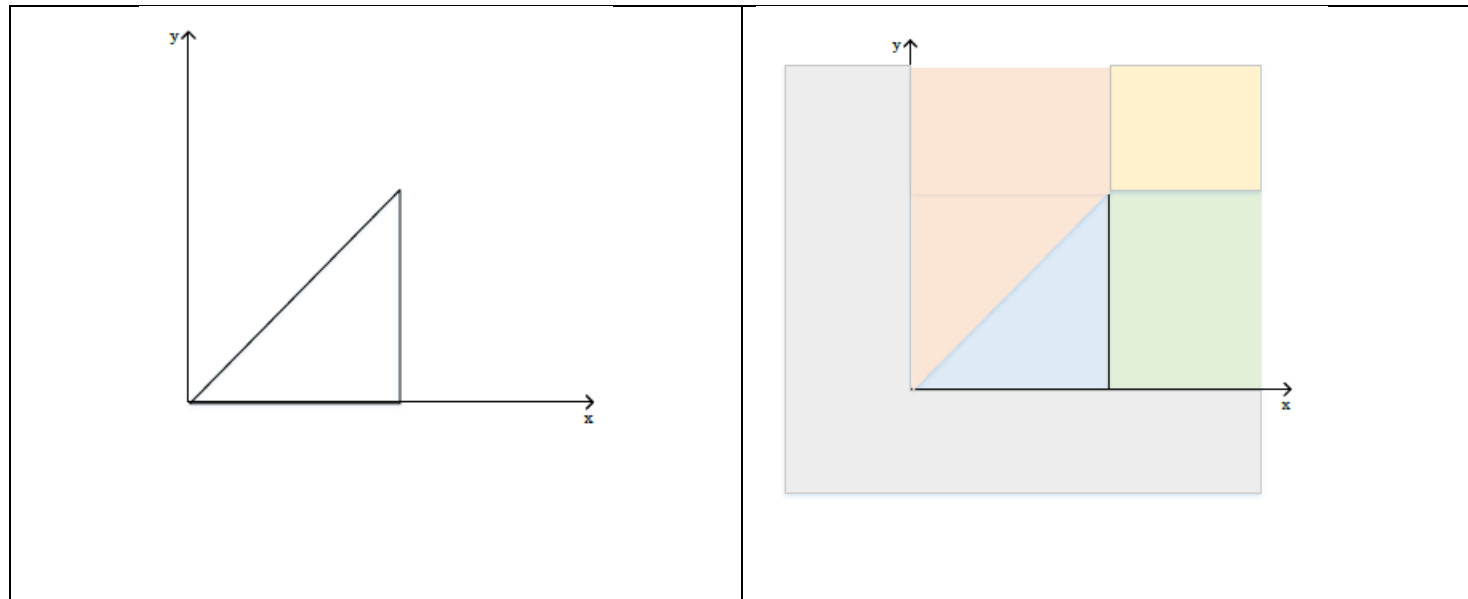


$$\int_0^y \int_v^x 2$$



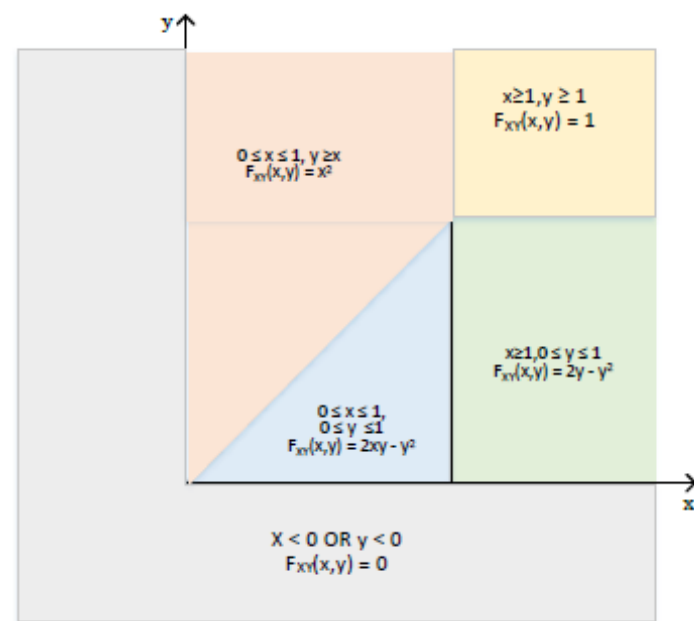
$$\int_0^x \int_0^y$$

$$\int_0^y \int_v^x 2 \, du \, dv$$

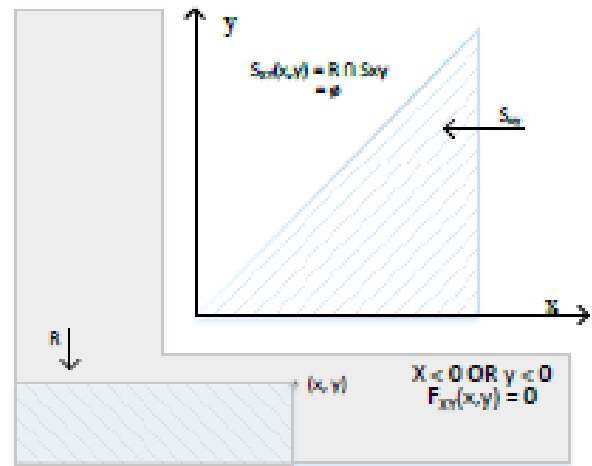


The joint pdf of X and Y $f_{XY}(x, y)$ is given by

$$f_{XY}(x, y) = \begin{cases} 2, & 0 \leq y \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

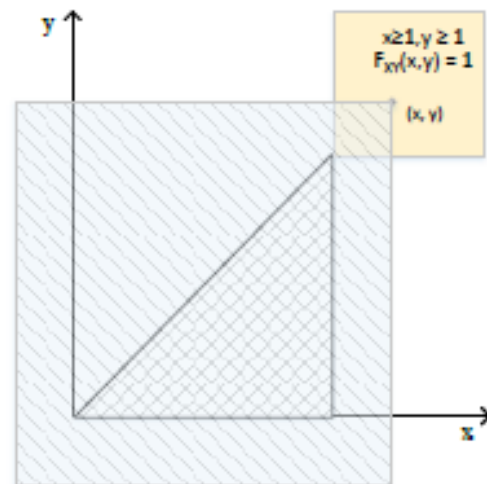


Case 1: $x < 0$ OR $y < 0$



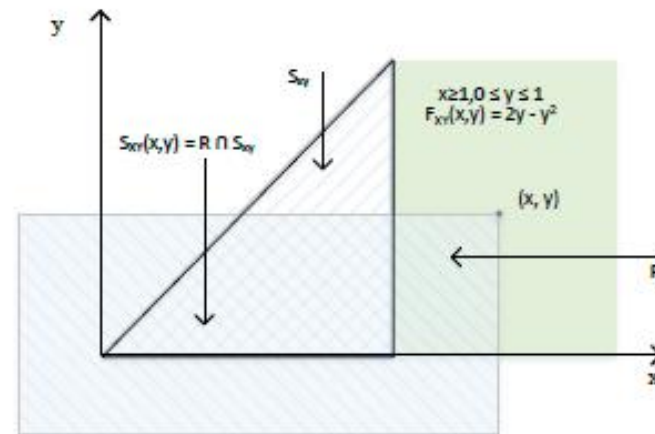
$$F_{XY}(x, y) = 0$$

Case 5: $x \geq 1, y \geq 1$



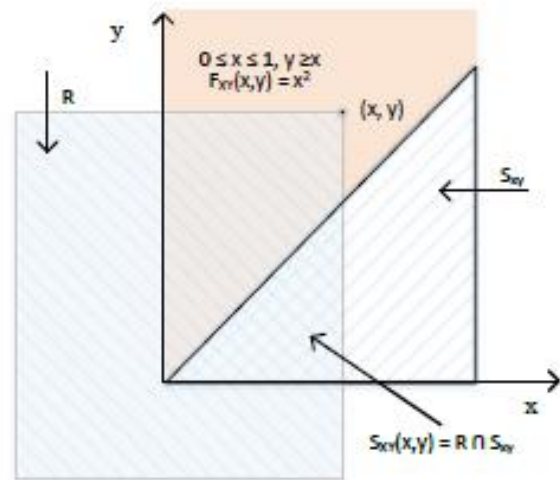
$$F_{XY}(x, y) = 1$$

Case 2: $x \geq 1, 0 \leq y < 1$



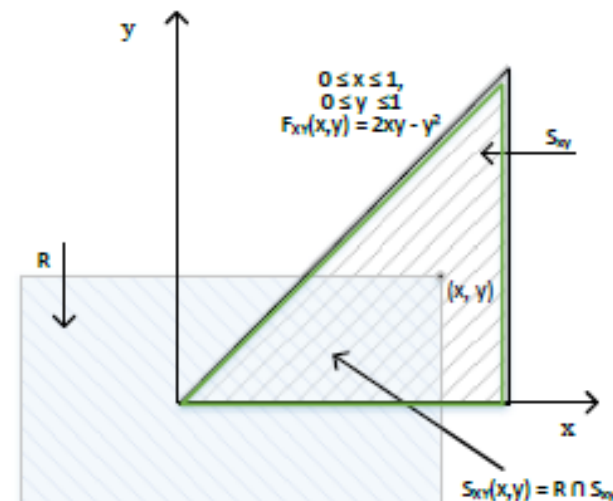
$$\begin{aligned} F_{X,Y}(x, y) &= \int_0^y \int_1^1 2dx dy \\ &= \int_0^y \left[2x \Big|_1^1 \right] dy \\ &= \int_0^y 2(1 - y) dy \\ &= \int_0^y 2dy - \int_0^y 2y dy \\ &= 2y \Big|_0^y - \frac{2y^2}{2} \Big|_0^y \\ &= 2y - y^2. \end{aligned}$$

Case 3: $0 \leq x < 1, y \geq x$



$$\begin{aligned}
 F_{X,Y}(x, y) &= \int_0^x \int_y^x 2dx dy \\
 &= \int_0^x \left[2x \Big|_y^x \right] dy \\
 &= \int_0^x (2x - 2y) dy \\
 &= 2xy \Big|_0^x - \frac{2y^2}{2} \Big|_0^x \\
 &= 2x^2 - x^2 \\
 &= x^2.
 \end{aligned}$$

Case 4: $0 \leq x < 1, 0 \leq y < 1$



$$\begin{aligned}
 F_{X,Y}(x, y) &= \int_0^y \int_y^x 2dx dy \\
 &= \int_0^y \left[2x \Big|_y^x \right] dy \\
 &= \int_0^y (2x - 2y) dy \\
 &= 2xy \Big|_0^y - \frac{2y^2}{2} \Big|_0^y \\
 &= 2xy - y^2.
 \end{aligned}$$

The complete JCDF of the two RVs is

$$F_{X,Y}(x,y) = \begin{cases} 0, & \text{for } x < 0 \text{ or } y < 0; \\ 2y - y^2, & \text{for } x > 1, 0 \leq y \leq 1; \\ x^2, & \text{for } 0 \leq x \leq 1, y \geq x; \\ 2xy - y^2, & \text{for } 0 \leq y \leq x \leq 1; \\ 1, & \text{for } x > 1, y > 1. \end{cases}$$

Marginal probability density function

Definition 4.9. *Marginal Probability Density Function.* If X and Y are random variable with joint PDF $f_{X,Y}(x, y)$, then the marginal PDF of X and Y are given by

$$f_X(x) = \int_{-\infty}^{\infty} f_{x,y}(x, y) dy, \quad (4.28)$$

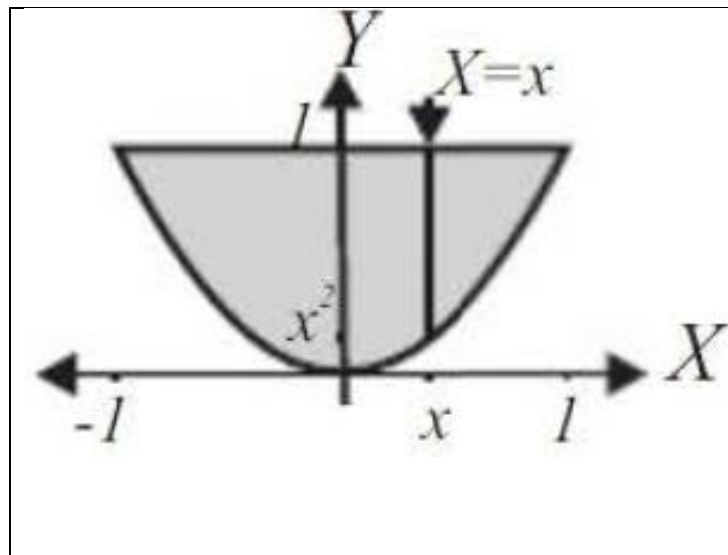
and

$$f_Y(y) = \int_{-\infty}^{\infty} f_{x,y}(x, y) dx. \quad (4.29)$$

Example 4.8. The joint PDF of X and Y is given by

$$f_{X,Y}(x,y) = \begin{cases} \frac{5}{4}y, & \text{for } -1 \leq x \leq 1, x^2 \leq y \leq 1; \\ 0, & \text{otherwise.} \end{cases}$$

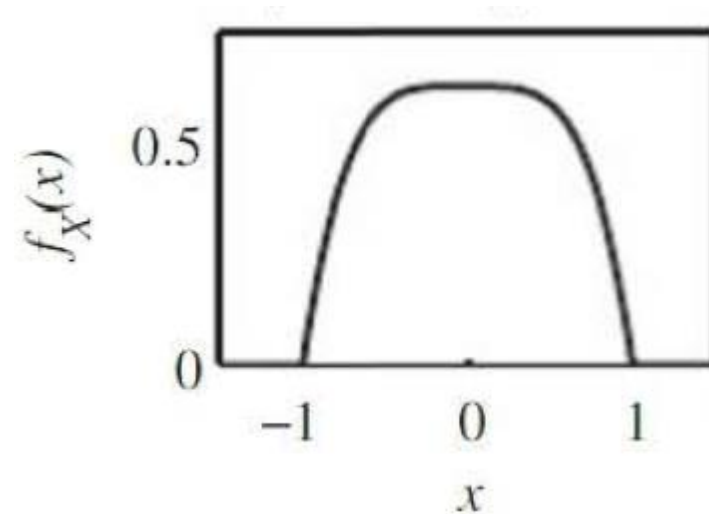
Find the marginal PDFs of X and Y .

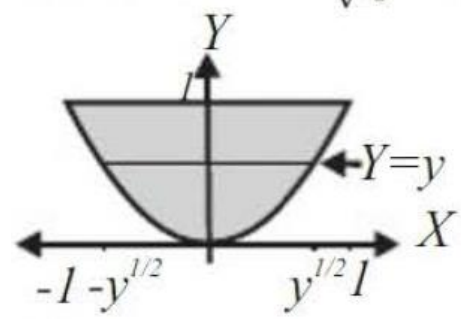


$$\begin{aligned}
 f_X(x) &= \int_{x^2}^1 f_{X,Y}(x,y) dy \\
 &= \int_{x^2}^1 \frac{5}{4} y dy \\
 &= \frac{5}{8} y^2 \Big|_{x^2}^1 \\
 &= \frac{5}{8} (1 - x^4).
 \end{aligned}$$

The complete expression for marginal PDF of X is

$$f_X(x) = \begin{cases} \frac{5}{8}(1 - x^4), & \text{for } -1 \leq x \leq 1; \\ 0, & \text{otherwise.} \end{cases}$$

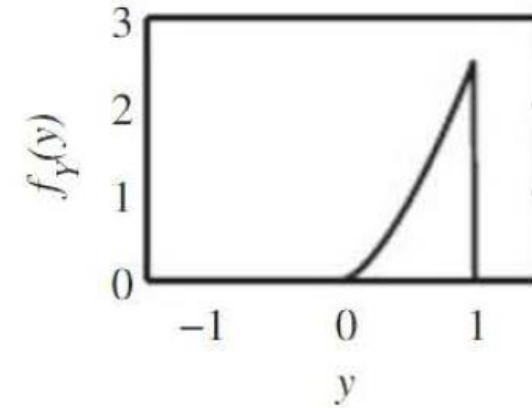




$$\begin{aligned}
 f_Y(y) &= \int_{-\sqrt{y}}^{+\sqrt{y}} \frac{5}{4} y dx \\
 &= \frac{5}{4} yx \Big|_{-\sqrt{y}}^{+\sqrt{y}} \\
 &= \frac{5}{4} y(\sqrt{y} + \sqrt{y}) \\
 &= \frac{5}{2} \cdot y \cdot y^{\frac{1}{2}} \\
 &= \frac{5}{2} \cdot y^{\frac{3}{2}}.
 \end{aligned}$$

The complete expression of the marginal PDF of Y is given by

$$f_Y(y) = \begin{cases} \frac{5}{2} y^{\frac{3}{2}}, & \text{for } 0 \leq y \leq 1; \\ 0, & \text{otherwise.} \end{cases}$$



Example 4.9. The joint PDF of random variables X and Y is given by

$$f_{X,Y}(x,y) = \begin{cases} \lambda xy^2, & \text{for } 0 \leq x \leq y \leq 1; \\ 0, & \text{otherwise.} \end{cases}$$

1. Determine the value of λ .
2. Find the marginal probability density function of X and Y .
3. Find $E[X]$ and $E[Y]$.

Value of λ

To find λ , note that $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy = 1$ gives

$$\int_0^1 \left(\int_x^1 \lambda xy^2 \, dy \right) dx = 1.$$

Therefore,

$$\begin{aligned} 1 &= \int_0^1 \left(\int_x^1 y^2 \, dy \right) \lambda x \, dx = \int_0^1 \left[\frac{1}{3} y^3 \right]_x^1 \lambda x \, dx \\ &= \lambda \int_0^1 \left(\frac{1}{3} - \frac{1}{3} x^3 \right) x \, dx = \frac{\lambda}{3} \int_0^1 (1 - x^3) x \, dx = \frac{\lambda}{3} \int_0^1 (x - x^4) \, dx \\ &= \frac{\lambda}{3} \left[\frac{1}{2} x^2 - \frac{1}{5} x^5 \right]_0^1 = \frac{\lambda}{10}, \end{aligned}$$

and hence $\lambda = 10$.

Marginal PDF of X

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) \, dy = \int_x^1 10xy^2 \, dy = \left[\frac{10}{3} xy^3 \right]_x^1 = \frac{10}{3} x(1 - x^3), \quad 0 \leq x \leq 1$$

Marginal PDF of Y

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) \, dx = \int_0^y 10xy^2 \, dx = \left[5x^2y^2 \right]_0^y = 5y^4, \quad 0 \leq y \leq 1$$

Multiple Random Variables: Multivariate Joint Distribution

- Random Experiments with more than two RVs
- Outcomes of the experiments are represented by points on an n dimensional space

The probability models associated are

1. Multivariate joint PMF (MJPMF) – represents the probability of occurring an outcome $x_1, \dots, x_n$ – simultaneously n RVs have the values (as real numbers)
$$X_1 = x_1, \dots, X_n = x_n$$
2. Multivariate Joint PDF (MJPDF) – represents the probability density at a point $(x_1, \dots, x_n)$ in the n -dimensional space
3. Multivariate Joint CDF (MJCDF) – represents the total probability of an n -dimensional space associated with the point $(x_1, \dots, x_n)$ and defined by the lines $X_1 = x_1, \dots, X_n = x_n$.

Marginal PMFs/JPMFs/MJPMFs, Marginal PDFs/JPDFs/MJPDFs and marginal CDFs/JCDFs/MJCDFs of all the subsets of RVs

Multiple Discrete RVs: Multivariate Joint Discrete Distributions

Multivariate Joint Probability Mass Function (MJPMF)

$$P_{X_1, \dots, X_n}(x_1, \dots, x_n) = P[X_1 = x_1, \dots, X_n = x_n].$$

Properties:

1. $P_{X_1, \dots, X_n}(x_1, \dots, x_n) \geq 0$.
2. If for some i , $1 \leq i \leq n$, $x_i \notin S_{X_i}$, then

$$P_{X_1, \dots, X_n}(x_1, \dots, x_n) = 0.$$

3. $\sum_{x_i \in S_{X_i}, 1 \leq i \leq n} P_{X_1, \dots, X_n}(x_1, \dots, x_n) = 1$.

Multivariate Joint CDF

$$\begin{aligned} F_{X_1,\dots,X_n}(x_1,\dots,x_n) &= P[X_1 \leq x_1,\dots,X_n \leq x_n], \quad x_i < +\infty, 1 \leq i \leq n \\ &= \sum_{x_1 \in S_{X+1}} \sum_{x_j \in S_{X_j}}^{x_i} P_{X_1,\dots,X_n}(x_1,\dots,x_n) \end{aligned}$$

Lower part need to be corrected

Marginal PMF for multivariate distributions

1. Marginal PMF
2. Marginal Joint PMF
3. Marginal Multivariate Joint PMF

Marginal PMF of X_i

$$\begin{aligned} P_{X_i}(x_i) &= P[X_i = x_i] \\ &= P[X_i = x_i, x_j \in S_{X_j}], \quad i \leq j \leq n, i \neq j \\ &= \sum_{x_1 \in S_{X_1}} \cdots \sum_{x_{i-1} \in S_{X_{i-1}}} \sum_{x_{i+1} \in S_{X_{i+1}}} \cdots \sum_{x_n \in S_{X_n}} P_{X_1, \dots, X_n}(x_1, \dots, x_n), \quad 1 \leq i \leq n \end{aligned}$$

Marginal Joint PMF of X_j and X_k

Let I is the set of indices of the RVs, $I = \{1, 2, \dots, n\}$

The marginal Joint PMF of any pair of RVS with indices $J = \{j, k\}, J \subset I$

$$\begin{aligned} P_{X_j X_k}(x_j, x_k) &= P[X_j = x_j, X_k = x_k] \\ &= \sum_{x_{i_1} \in S_{X_{i_1}}} \dots \sum_{x_{i_{n-2}} \in S_{X_{i_{n-2}}}} P_{X_1, \dots, X_n}(x_1, \dots, x_n), \quad i_1, \dots, i_{n-2} \in I, i_1, \dots, i_{n-2} \notin J \end{aligned}$$

The Marginal Multivariate Joint PMF of any subset of RVS from I with indices $J = \{k_1, \dots, k_m\}, J \subset I$ and $m < n$

$$\begin{aligned} P_{X_j X_k}(x_{k_1}, \dots, x_{k_m}) &= P[X_{k_1} = x_{k_1}, \dots, X_{k_m} = x_{k_m}] \\ &= \sum_{x_{i_1} \in S_{X_{i_1}}} \dots \sum_{x_{i_{n-m}} \in S_{X_{i_{n-m}}}} P_{X_1, \dots, X_n}(x_1, \dots, x_n), \quad i_1, \dots, i_{n-m} \in I, i_1, \dots, i_{n-m} \notin J \end{aligned}$$

Marginal CDF multiple RVs

$$\begin{aligned} F_{X_i}(x) &= P[X_i \leq x] \\ &= P[X_1 < \infty, \dots, X_{i-1} < \infty, X_i = x, X_{i+1} < \infty, X_n < \infty] \\ &= \sum_{x_1 < \infty} \cdots \sum_{x_{i-1} < \infty} \sum_{x_i < x} \sum_{x_{i+1} < \infty} \cdots \sum_{x_n < \infty} P_{X_1, \dots, X_n}(x_1, \dots, x_n) \\ &= \lim_{\substack{x_j \rightarrow \infty \\ 1 \leq j \leq n, j \neq i}} F_{X_1, \dots, X_n}(x_1, \dots, x_n). \end{aligned}$$

Example: In a box, there are 23 balls of which 5 balls are red, 8 balls are blue and 10 balls are green. Suppose 7 balls are picked from the box without replacement. Let X , Y and Z represent the number of red, blue and green balls picked. Find the multivariate joint PMF and marginal PMFs of X , Y and Z .

Let the multivariate joint PMF is denoted by $P_{XYZ}(x, y, z)$

$$P_{XYZ}(x, y, z) = \frac{\text{\textit{\# of ways exactly } } x \text{ red, } y \text{ blue and } z \text{ green balls picked}}{\text{\textit{\# of ways 7 balls are picked from 23 balls}}}$$
$$= \frac{a}{b}$$

The probability represented by $P_{XYZ}(x, y, z)$ is the ratio of a to b , where:

1. a counts the number of ways x red, y blue and z green balls can be picked which is the product of three terms p, q , and r , $a = pqr$, where

- a. p counts the number of ways x red balls are picked from 5 red balls,

$$p = \binom{5}{x}$$

- b. q counts the number of ways y blue balls are picked from 8 blue

$$\text{balls, } q = \binom{8}{y}$$

- c. r counts the number of ways z green balls are picked from 10 green

$$\text{balls, } r = \binom{10}{z}$$

$$a = pqr = \binom{5}{x} \binom{8}{y} \binom{10}{z}$$

2. b counts the number of ways 7 balls are picked from 23 balls, $b = \binom{23}{7}$

Therefore, the MJPMF of X, Y and Z $P_{X,Y,Z}(x, y, z)$, for $0 \leq x \leq 5, 0 \leq y \leq 8, 0 \leq z \leq 10$, and $x + y + z = 7$, can be given by

$$P_{X,Y,Z}(x, y, z) = \frac{\binom{5}{x} \binom{8}{y} \binom{10}{z}}{\binom{23}{7}}$$

and $P_{X,Y,Z}(x, y, z) = 0$ otherwise.

$$P_{XYZ}(x, y, z) = \begin{cases} \frac{\binom{5}{x}\binom{8}{y}\binom{10}{z}}{\binom{23}{7}}, & \begin{matrix} 0 \leq x \leq 5 \\ 0 \leq y \leq 8 \\ 0 \leq z \leq 10 \\ x + y + z = 7 \end{matrix} \\ 0, & \text{otherwise} \end{cases}$$

Marginal PMFs of X , Y and Z

The marginal PMF of X , $P_X(x)$, can be obtained in the following way:

$$\begin{aligned} P_X(x) &= P[X = x] \\ &= \sum_{\substack{x+y+z=7 \\ 0 \leq y \leq 7 \\ 0 \leq z \leq 7}} P_{X,Y,Z}(x,y,z) \\ &= \sum_{y=0}^{7-x} \frac{\binom{5}{x} \binom{8}{y} \binom{10}{7-x-y}}{\binom{23}{7}} \\ &= \frac{\binom{5}{x}}{\binom{23}{7}} \sum_{y=0}^{7-x} \binom{8}{y} \binom{10}{7-x-y}. \end{aligned}$$

$\sum_{y=0}^{7-x} \binom{8}{y} \binom{10}{7-x-y}$ is the total number of ways $(7 - x)$ blue or green balls can be picked from 18 balls

Which is, number of ways $(7 - x)$ balls are picked from 18 balls and is $\binom{18}{7-x}$

Therefore, the marginal PMF of X is

$$P_X(x) = \frac{\binom{5}{x} \binom{18}{7-x}}{\binom{23}{7}}.$$

Similarly,

$$P_Y(y) = \frac{\binom{8}{y} \binom{18}{7-y}}{\binom{23}{7}}$$

$$P_Z(z) = \frac{\binom{10}{z} \binom{18}{7-z}}{\binom{23}{7}}.$$

Multivariate Continuous Random Variables

Multivariate Joint CDF (MJCDF)

$$F_{X_1, \dots, X_n}(x_1, \dots, x_n) = P[X_1 \leq x_1, \dots, X_n \leq x_n]$$

If the joint distribution of n (where $n > 2$) random variables $X_1, \dots, X_n$ is continuous, then the multivariate joint PDF, $f_{X_1, \dots, X_n}(x_1, \dots, x_n)$, can be derived from the MJCDF using the following relation

$$f_{X_1, \dots, X_n}(x_1, \dots, x_n) = \frac{\partial^n F_{X_1, \dots, X_n}(x_1, \dots, x_n)}{\partial x_1, \dots, \partial x_n},$$

$$F_{X_1, \dots, X_n}(x_1, \dots, x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f_{X_1, \dots, X_n}(x_1, \dots, x_n) dx_1 \dots dx_n.$$

Marginal density function for multiple RVs

If the multivariate joint PDF of $X_1, \dots, X_n$ is $f_{X_1, \dots, X_n}(x_1, \dots, x_n)$, then the marginal PDF of X_1 , $f_{X_1}(x_1)$, can be given by

$$f_{X_1}(x_1) = \underbrace{\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty}}_{n-1} f_{X_1, \dots, X_n}(x_1, \dots, x_n) dx_2 \dots dx_n.$$

If the multivariate Joint PDF of 4 random variables $X_1, X_2, \dots, X_4$ is $f_{X_1, \dots, X_4}(x_1, \dots, x_4)$, then

The marginal PDF of X_1 , $f_{X_1}(x_1)$, is given by

$$f_{X_1}(x_1) = \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} f_{X_1, \dots, X_4}(x_1, \dots, x_4) \, dx_2 dx_3 dx_4$$

The marginal joint PDF of X_1 and X_2 , $f_{X_1 X_2}(x_1, x_2)$, is given by

$$f_{X_1 X_2}(x_1, x_2) = \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} f_{X_1, \dots, X_4}(x_1, \dots, x_4) \, dx_3 dx_4$$

The marginal multivariate joint PDF X_1 , X_2 and X_3 , $f_{X_1, \dots, X_3}(x_1, \dots, x_3)$ is given by

$$f_{X_1, \dots, X_3}(x_1, \dots, x_3) = \int_{-\infty}^{+\infty} f_{X_1, \dots, X_4}(x_1, \dots, x_4) \, dx_4$$

Example 5.2. Suppose that a machine has three parts, and part i will fail at time X_i for $i = 1, 2, 3$. The following function may be the multivariate joint CDF of X_1, X_2 and X_3 .

$$F_{X_1, X_2, X_3}(x_1, x_2, x_3) = \begin{cases} (1 - e^{-x_1})(1 - e^{-2x_2})(1 - e^{-3x_3}), & \text{for } x_1, x_2, x_3 \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

- a) Find the multivariate joint PDF of X_1, X_2, X_3 .
- b) Find the marginal joint PDF of X_1 and X_2 .

a) The multivariate joint PDF of X_1, X_2, X_3 is given by

$$\begin{aligned} f_{X_1, X_2, X_3}(x_1, x_2, x_3) &= \frac{\partial^3 F_{X_1, X_2, X_3}(x_1, x_2, x_3)}{\partial x_1 \partial x_2 \partial x_3} \\ &= e^{x_1} . 2e^{2x_2} . 3e^{3x_3} \\ &= 6e^{-x_1-2x_2-3x_3}, \quad x_1, x_2, x_3 \geq 0. \end{aligned}$$

Failure Times. We can find the marginal bivariate c.d.f. of X_1 and X_3 from the joint c.d.f. in Example 3.7.2 by letting x_2 go to ∞ . The limit is

$$F_{13}(x_1, x_3) = \begin{cases} (1 - e^{-x_1})(1 - e^{-3x_3}) & \text{for } x_1, x_3 \geq 0, \\ 0 & \text{otherwise.} \end{cases} \blacktriangleleft$$

Example: Consider that three continuous RVs X, Y and Z are jointly uniformly distributed in the region $0 \leq z \leq y \leq x \leq 1$.

1. Find the multivariate joint PDF $f_{XYZ}(x, y, z)$.
2. Find the marginal joint PDFs $f_{XY}(x, y)$, $f_{XZ}(x, z)$ and $f_{YZ}(y, z)$.
3. Find the marginal PDFs $f_X(x)$, $f_Y(y)$ and $f_Z(z)$,

Since $f_{XYZ}(x, y, z)$ is jointly continuous, $f_{XYZ}(x, y, z) = c$

Multivariate joint PDF	
$\int_0^1 \int_0^x \int_0^y c \, dz \, dy \, dx = 1$ $= \int_0^1 \int_0^x (cz _0^y) \, dy \, dx = \int_0^1 \frac{cy^2}{2} \Big _0^x \, dx$ $= \int_0^1 \frac{1}{2} cx^2 \, dx = \frac{1}{2} c \frac{x^3}{3} \Big _0^1 = \frac{c}{6}$ $\Rightarrow c = 6$	

Marginal joint PDF of any pair of X, Y and Z

$0 \leq z \leq y \leq x \leq 1$			
RVS	z	y	x
Limits	$0 \leq z \leq y$	$z \leq y \leq x$	$y \leq x \leq 1$

Marginal Joint PDF $f_{XY}(x, y)$		
PDF	$f_{XY}(x, y) = \int_0^y 6 \, dz$ $= 6z \Big _0^y$ $= 6y$	$f_{XY}(x, y) = \begin{cases} 6y, & 0 \leq y \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$
Proof	$\int_0^1 \int_0^x 6y \, dy \, dx = \int_0^1 \left(6 \frac{y^2}{2} \Big _0^x \right) dx = \int_0^1 3x^2 \, dx = 3 \frac{x^3}{3} \Big _0^1 = 1$	

Marginal Joint PDF $f_{XZ}(x, z)$		
PDF	$f_{XZ}(x, z) = \int_z^x 6 \, dy$ $= 6y \Big _z^x$ $= 6(x - z)$	$f_{XZ}(x, z) = \begin{cases} 6(x - z), & 0 \leq z \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$
Proof	$\int_0^1 \int_0^x 6(x - z) \, dz \, dx = \int_0^1 \left(6xz \Big _0^x - 6 \frac{z^2}{2} \Big _0^x \right) dx = \int_0^1 (6x^2 - 3x^2) \, dx$ $= 3 \frac{x^3}{3} \Big _0^1 = 1$	

Marginal Joint PDF $f_{YZ}(y, z)$		
PDF	$f_{YZ}(y, z) = \int_y^1 6 \, dx$ $= 6x \Big _y^1$ $= 6(1 - y)$	$f_{YZ}(y, z) = \begin{cases} 6(1 - y), & 0 \leq z \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$
Proof	$\int_0^1 \int_0^y 6(1 - y) \, dz \, dy = \int_0^1 (6z \Big _0^y - 6yz \Big _0^y) \, dy = \int_0^1 (6y - 6y^2) \, dy$ $= 6 \frac{y^2}{2} \Big _0^1 - 6 \frac{y^3}{3} \Big _0^1 = 3 - 2 = 1$	

Marginal PDF of X	Marginal joint PDF of X, Y and X, Z
$f_X(x) = \int_0^x 6y \, dy = 6 \frac{y^2}{2} \Big _0^x = 3x^2$	$f_{XY}(x, y) = \begin{cases} 6y, & 0 \leq y \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$
$f_X(x) = \int_0^x 6(x - z) \, dz$ $= 6xz \Big _0^x - 6 \frac{z^2}{2} \Big _0^x = 6x^2 - 3x^2$ $= 3x^2$	$f_{XZ}(x, z) = \begin{cases} 6(x - z), & 0 \leq z \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$
$f_X(x) = \begin{cases} 3x^2, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$	

Marginal PDF of Y	Marginal joint PDF of X, Y and Y, Z
$f_Y(y) = \int_y^1 6y \, dx = 6yx _y^1 =$ $= 6y - 6y^2 = 6y(1 - y)$	$f_{XY}(x, y) = \begin{cases} 6y, & 0 \leq y \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$
$f_Y(y) = \int_0^y 6(1 - y) \, dz$ $= 6z _0^y - 6yz _0^y = 6y - 6y^2$ $= 6y(1 - y)$	$f_{YZ}(y, z) = \begin{cases} 6(1 - y), & 0 \leq z \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$
$f_Y(y) = \begin{cases} 6y(1 - y), & 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$	

Marginal PDF of Z	Marginal joint PDF of X, Z and Y, Z
$f_Y(y) = \int_y^1 6y \, dx = 6yx _y^1 =$ $= 6y - 6y^2 = 6y(1 - y)$	$f_{XY}(x, y) = \begin{cases} 6y, & 0 \leq y \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$
$f_Y(y) = \int_0^y 6(1 - y) \, dz$ $= 6z _0^y - 6yz _0^y = 6y - 6y^2$ $= 6y(1 - y)$	$f_{YZ}(y, z) = \begin{cases} 6(1 - y), & 0 \leq z \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$
$f_Y(y) = \begin{cases} 6y(1 - y), & 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$	

Random Variables $Y_1, \dots, Y_4$ have the joint PDF

$$f_{Y_1, \dots, Y_4}(y_1, \dots, y_4) = \begin{cases} 4 & 0 \leq y_1 \leq y_2 \leq 1, 0 \leq y_3 \leq y_4 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the marginal PDFs $f_{Y_1, Y_4}(y_1, y_4)$, $f_{Y_2, Y_3}(y_2, y_3)$ and $f_{Y_3}(y_3)$.

For $0 \leq y_1 \leq 1$ and $0 \leq y_4 \leq 1$

$$f_{Y_1, Y_4}(y_1, y_4) = \int_{y_1}^1 \int_0^{y_4} 4 \, dy_3 \, dy_2 = 4(1 - y_1)y_4.$$

The complete expression for $f_{Y_1, Y_4}(y_1, y_4)$ is

$$f_{Y_1, Y_4}(y_1, y_4) = \begin{cases} 4(1 - y_1)y_4 & 0 \leq y_1 \leq 1, 0 \leq y_4 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Similarly, for $0 \leq y_2 \leq 1$ and $0 \leq y_3 \leq 1$,

$$f_{Y_2, Y_3}(y_2, y_3) = \int_0^{y_2} \int_{y_3}^1 4 \, dy_4 \, dy_1 = 4y_2(1 - y_3).$$

The complete expression for $f_{Y_2, Y_3}(y_2, y_3)$ is

$$f_{Y_2, Y_3}(y_2, y_3) = \begin{cases} 4y_2(1 - y_3) & 0 \leq y_2 \leq 1, 0 \leq y_3 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

For $0 \leq y_3 \leq 1$

$$f_{Y_3}(y_3) = \int_{-\infty}^{\infty} f_{Y_2, Y_3}(y_2, y_3) \, dy_2 = \int_0^1 4y_2(1 - y_3) \, dy_2 = 2(1 - y_3).$$

The complete expression is

$$f_{Y_3}(y_3) = \begin{cases} 2(1 - y_3) & 0 \leq y_3 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$